CIE AS PHYSICS (9702) • QUESTION PAPER

9.3 Resistance & Resistivity

Syllabus Section: 9. Electricity **Paper Type:** Multiple Choice (Paper 1) **Total:** 60 Questions (60 Marks)

Question 1 Easy [1 mark]

Which equation is used to define resistance?

A energy = (current)^2 × resistance × time

B potential difference = current × resistance

C power = (current)^2 × resistance

D resistivity = resistance × area ÷ length

Question 2 Easy [1 mark]

What is the unit of resistance when expressed in SI base units?

A kg m^2 s^{-2} A^{-1}

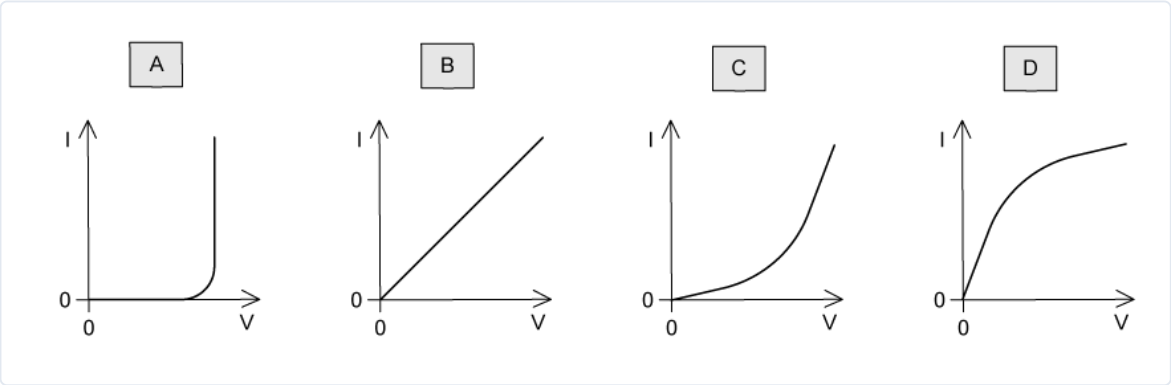
B kg m^2 s^{-3} A^{-2}

C kg m s^{-2} A^{-1}

D kg m s^{-3} A^{-1}

Question 3 Easy [1 mark]

Which graph shows the *I*–*V* characteristic of a filament lamp ?



A

B

C

D

Question 4 Easy [1 mark]

The resistance *R* of a resistor is determined by measuring the potential difference *V* across it and the current *I* in it. The value of *R* is then calculated using the equation

$$R = \frac{V}{I}$$

The values measured are $V = 1.00 \pm 0.05$ V and $I = 0.50 \pm 0.01$ A.

What is the percentage uncertainty in the value of *R* ?

A 2.5 %

B 3.0 %

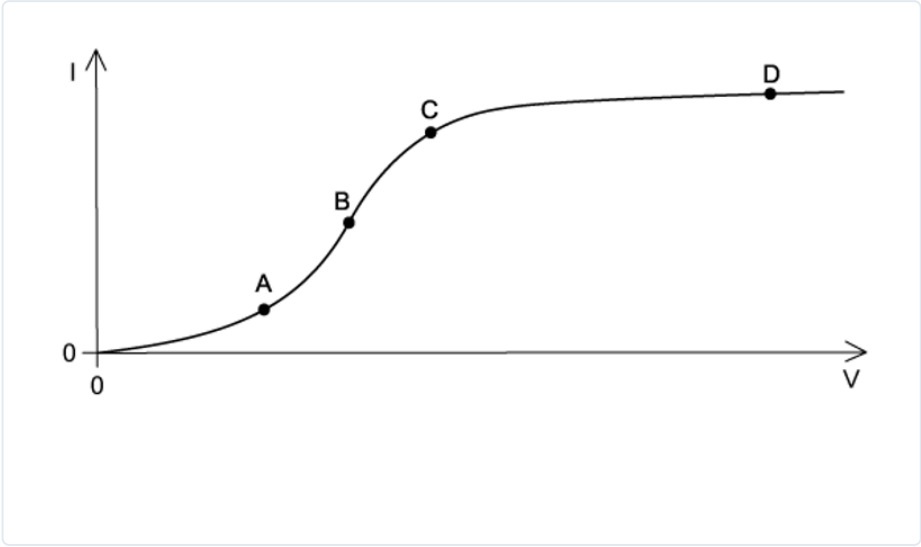
C 7.0 %

D 10.0 %

Question 5 Easy [1 mark]

The graph shows how the electric current *I* through a conducting liquid varies with the potential difference *V* across it.

At which point on the graph does the liquid have the smallest resistance?



A

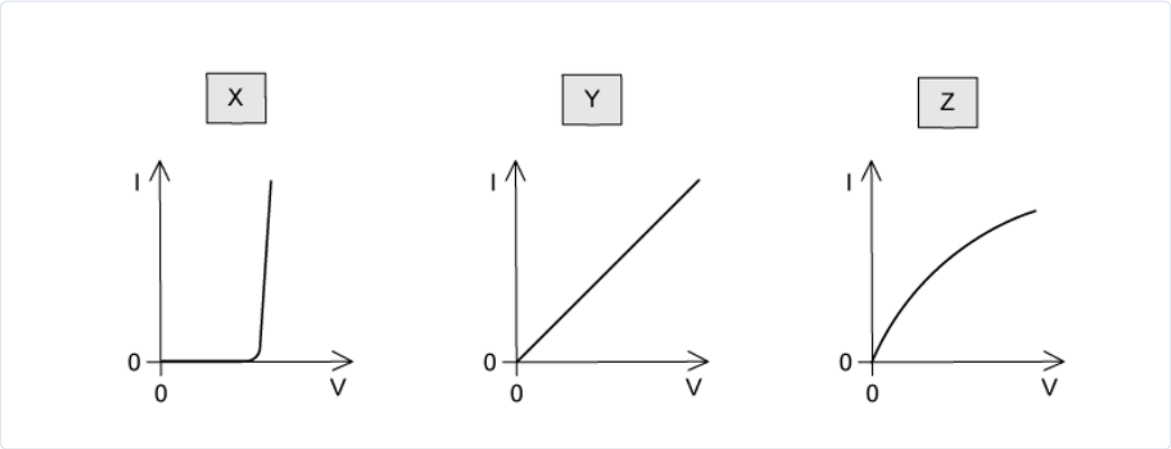
B

C

D

Question 6 Easy [1 mark]

The graphs show the variation with potential difference *V* of the current *I* for three circuit components.



The components are a metal wire at constant temperature, a semiconductor diode and a filament lamp.

Which row of the table correctly identifies these graphs?

	Metal wire at constant temperature	Semiconductor diode	Filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	X	Y

A

B

C

D

Question 7 Easy [1 mark]

What is a correct statement of Ohm's Law?

A the potential difference across a component equals the current providing the resistance and other physical conditions stay constant

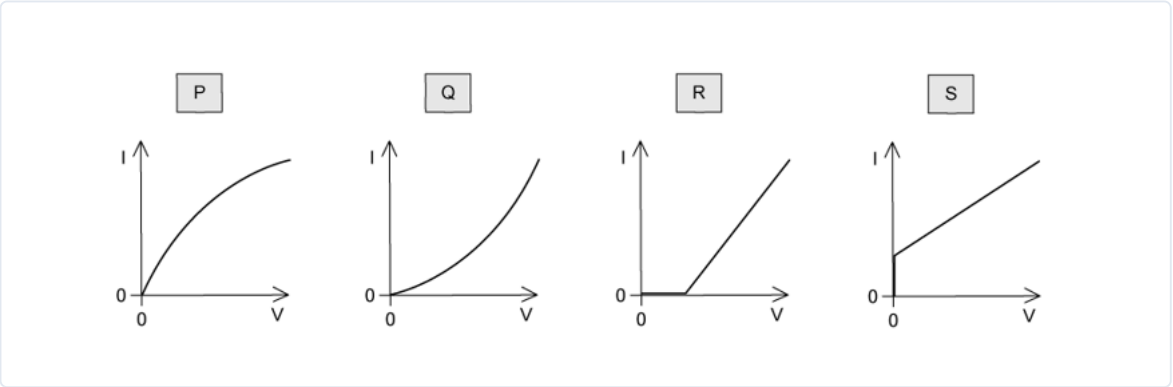
B the potential difference across a component equals the current multiplied by the resistance

C the potential difference across a component is proportional to its resistance.

D the potential difference across a component is proportional to the current in it providing physical conditions stay constant

Question 8 Easy [1 mark]

The graphs show possible current-voltage (*I*–*V*) relationships for a filament lamp and a semiconductor diode.



Which row best specifies the correct *I*–*V* graphs for the lamp and the diode?

	Filament lamp	Semiconductor diode
A	P	R
B	P	S
C	Q	R
D	Q	S

A

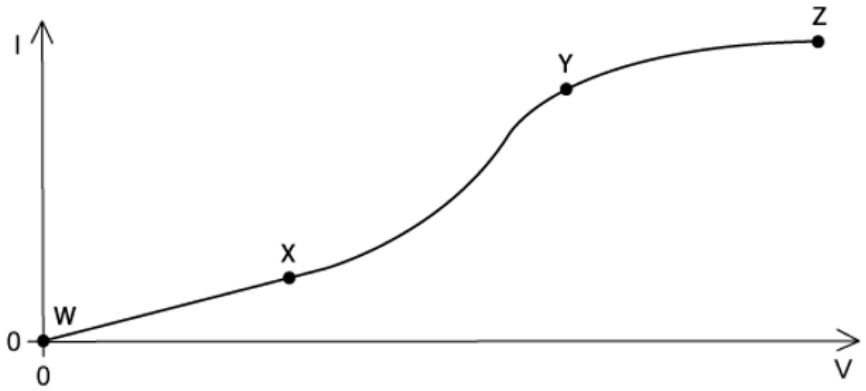
B

C

D

Question 9 Easy [1 mark]

An electrical component has a potential difference *V* across it and a current *I* through it. A graph of *I* against *V* is drawn and is marked in three sections WX, XY and YZ.



In which ways does the resistance of the component vary within each of the three sections?

	WX	XY	YZ
A	constant	decreases	increases
B	constant	increases	increases
C	increases	decreases	constant
D	increases	increases	decreases

A

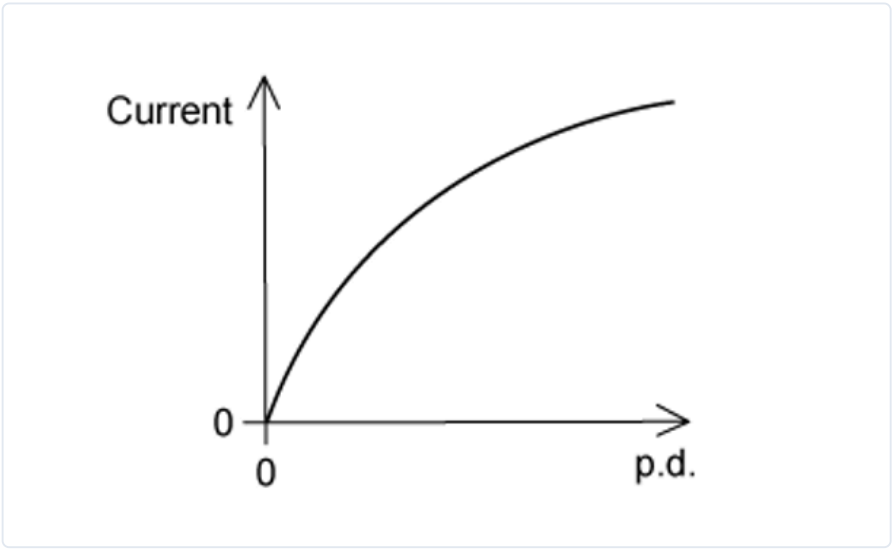
B

C

D

Question 10 Easy [1 mark]

The graph shows the variation with potential difference (p. d.) of the current in a lamp filament.



Which statement explains the shape of this graph?

A as the filament temperature rises, electrons can pass more easily through the filament

B it takes time for the filament to reach its working temperature

C the power output of the filament is proportional to the square of the current in it

D the resistance of the filament increases with a rise in temperature

Question 11 Easy [1 mark]

The unit of resistivity, expressed in terms of base units, is given by

$$\text{kg} \quad x^3 \quad y^{-2} \quad z^{-3}$$

Which base units are *x*, *y* and *z* ?

	x	*y*	*z*
A	ampere	metre	second
B	metre	ampere	second
C	metre	second	ampere
D	second	ampere	metre

A

B

C

D

Question 12 Easy [1 mark]

A cylindrical wire of length 10 m and diameter 2.0 mm has a resistance of 0.050 Ω.

From which material is the wire made?

	Material	Resistivity / Ω m
A	bronze	1.6×10^{-7}
B	nichrome	1.6×10^{-6}
C	silver	1.6×10^{-8}
D	zinc	6.3×10^{-8}

A

B

C

D

Question 13 Easy [1 mark]

The resistance of one device R_1 is designed to change with temperature; the resistance of another device R_2 is designed to change with illumination.

What are the names of these devices ?

	R_1	R_2
A	thermistor	light-dependent resistor
B	thermistor	potential divider
C	semiconductor diode	light-dependent resistor
D	semiconductor diode	potential divider

A

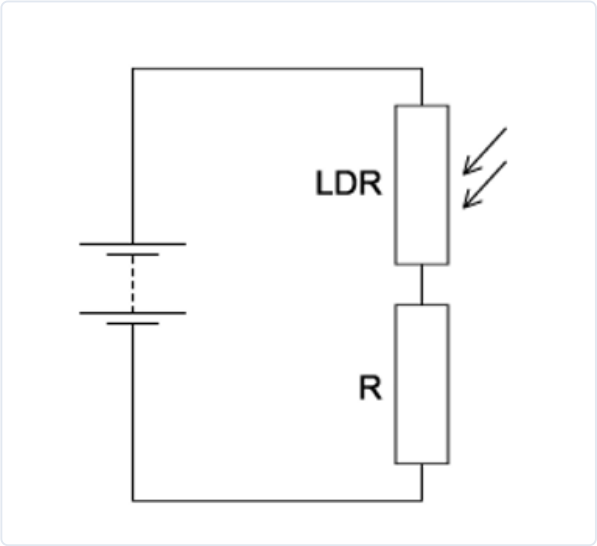
B

C

D

Question 14 Easy [1 mark]

A light-dependent resistor (LDR) is connected in series with a resistor *R* and a battery.



The resistance of the LDR is equal to the resistance of *R* when no light falls on the LDR. When the light intensity falling on the LDR increases, which statement is correct?

A the current in *R* decreases

B the current through the LDR decreases

C the p. d. across *R* decreases

D the p .d. across the LDR decreases

Question 15 Easy [1 mark]

A copper wire is stretched so that its diameter is reduced from 1.0 mm to a uniform 0.5 mm. The resistance of the unstretched copper wire is 0.2 Ω.

What is the resistance of the stretched wire?

A 0.4 Ω

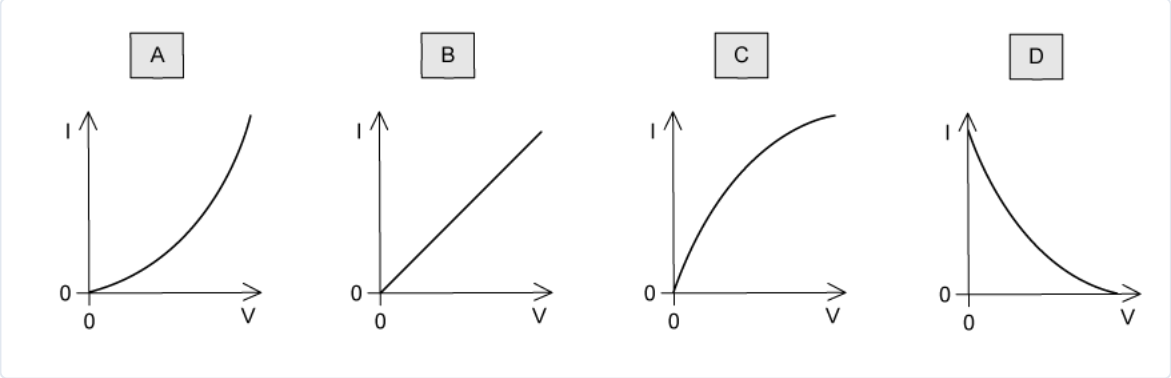
B 0.8 Ω

C 1.6 Ω

D 3.2 Ω

Question 16 Easy [1 mark]

Which graph best represents the way in which the current *I* through a thermistor depends upon the potential difference *V* across it ?



A

B

C

D

Question 17 Easy [1 mark]

A power cable has length 2000 m. The cable is made of twelve parallel strands of copper wire, each with diameter 0.51 mm.

What is the resistance of the cable ? (resistivity of copper = 1.7×10^{-8} Ω m)

A 0.014 Ω

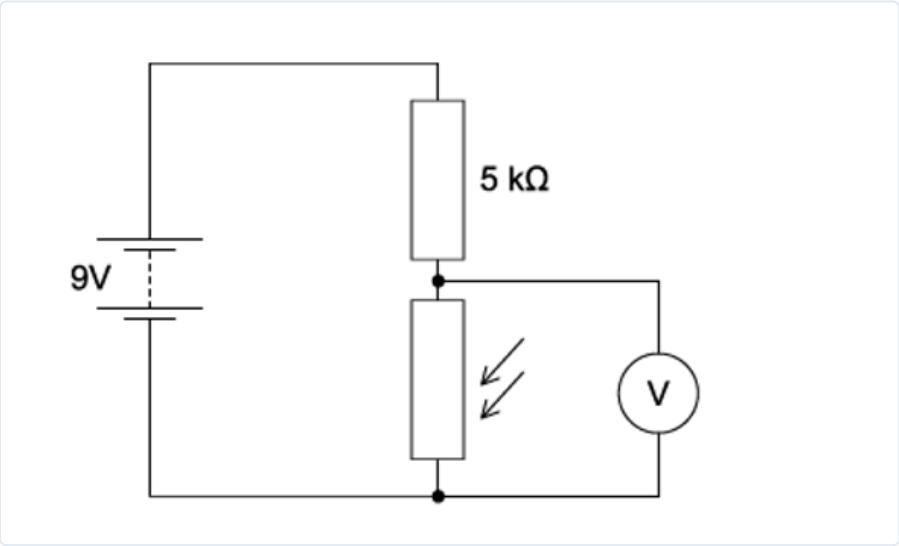
B 3.5 Ω

C 14 Ω

D 166 Ω

Question 18 Easy [1 mark]

A circuit is set up with an LDR and a fixed resistor as shown.



The voltmeter reads 4 V.

The light intensity is increased.

What is a possible voltmeter reading?

A 3V

B 4V

C 6V

D 8V

Question 19 Easy [1 mark]

A cylindrical wire 4.0 m long has a resistance of 31 Ω and is made of metal of resistivity 1.0×10^{-6} Ω m.

What is the radius of cross-section of the wire?

A 1.0×10^{-8} m

B 2.0×10^{-8} m

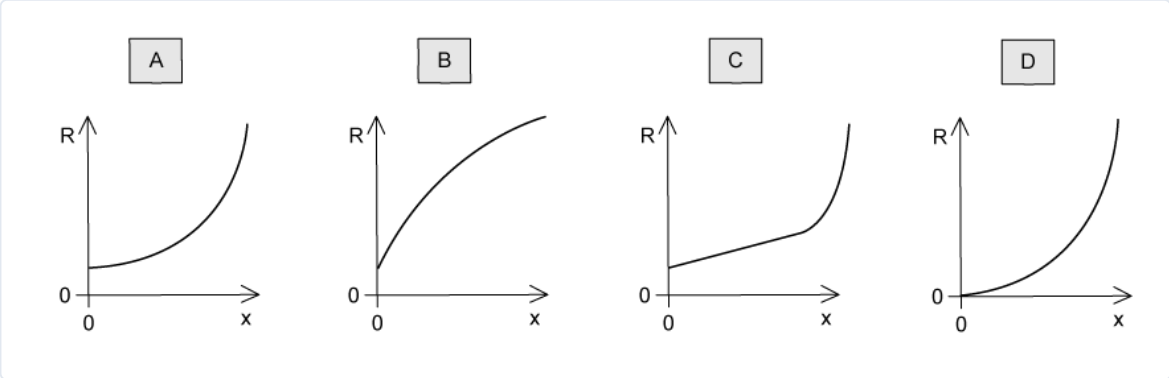
C 6.4×10^{-8} m

D 2.0×10^{-4} m

Question 20 Easy [1 mark]

When a thin metal wire is stretched, it becomes longer and thinner. This causes a change in the resistance of the wire. The volume of the wire remains constant.

Which graph could represent the variation with extension *x* of the resistance *R* of the wire ?



A

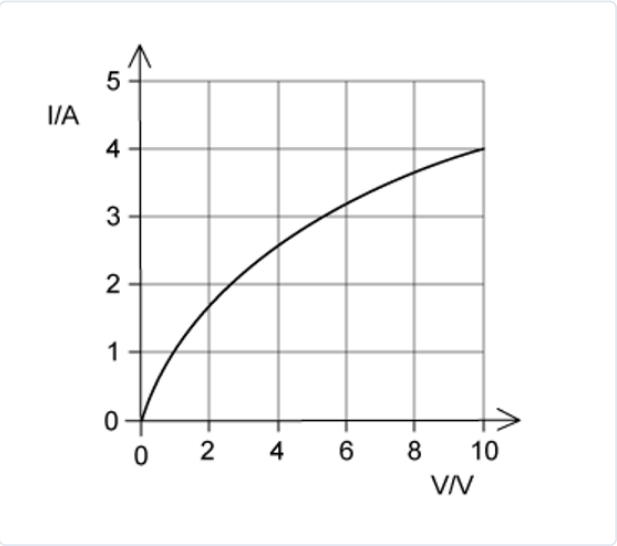
B

C

D

Question 21 Medium [1 mark]

The graph shows how current *I* varies with voltage *V* for a filament lamp.



Since the graph is not a straight line, the resistance of the lamp varies with *V*.

Which row gives the correct resistance at the stated value of *V*?

	***V* / V**	***R* / Ω**
A	2.0	1.5
B	4.0	3.2
C	6.0	1.9
D	8.0	0.9

A

B

C

D

Question 22 Medium [1 mark]

A charge of 8.0 C passes through a resistor of resistance 30 Ω at a constant rate in a time of 20 s.

What is the potential difference across the resistor?

A 0.40 V

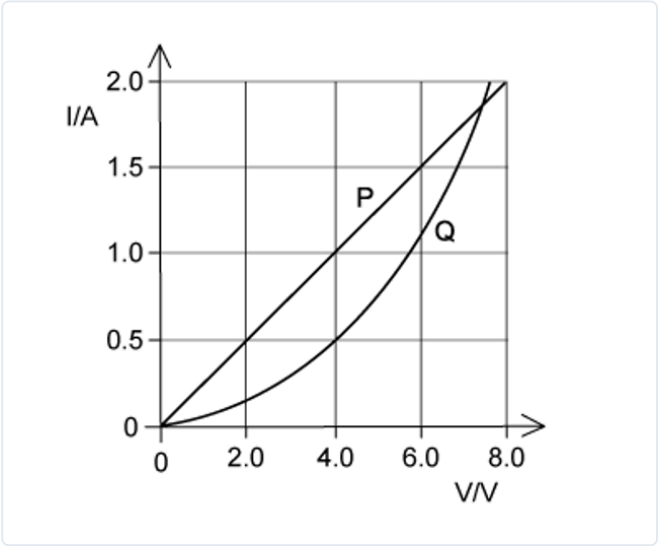
B 5.3 V

C 12 V

D 75 V

Question 23 Medium [1 mark]

The *I* -*V* characteristics of two electrical components P and Q are shown below.



Which statement is correct ?

A P is a resistor and Q is a filament lamp

B the resistance of Q increases as the current in it increases

C for a current of 1.9 A, the resistance of Q is approximately half that of P

D for a current of 0.5 A, the power dissipated in Q is double that in P

Question 24 Medium [1 mark]

The resistance R of a resistor is to be determined. The current I in the resistor and the potential difference V across it are measured.

The results, with their uncertainties, are

$I = (2.0 \pm 0.2)$ A $V = (15.0 \pm 0.5)$ V

What is the uncertainty in this value for R ?

A ± 0.3 Ω

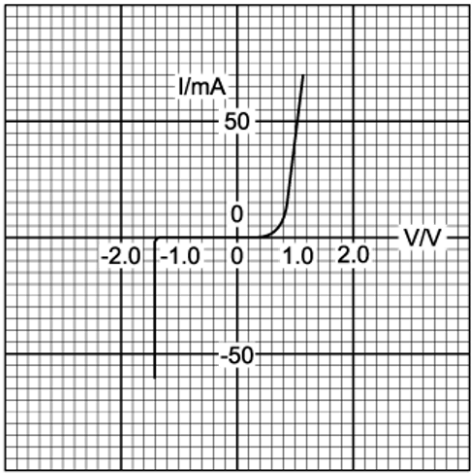
B ± 0.5 Ω

C ± 0.7 Ω

D ± 1 Ω

Question 25 Medium [1 mark]

The variation with potential difference *V* of the current *I* in a semiconductor diode is shown below.



What is the resistance of the diode for applied potential differences of +1.0 V and –1.0 V ?

	Resistance	
	at +1.0 V	**at –1.0 V**
A	20 Ω	infinite
B	20 Ω	zero
C	0.05 Ω	infinite
D	0.05 Ω	zero

A

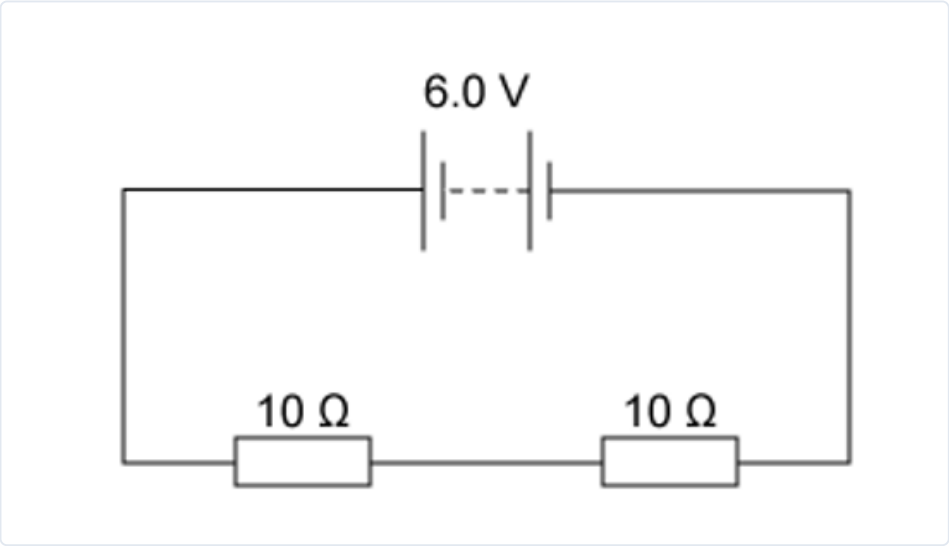
B

C

D

Question 26 Medium [1 mark]

A battery of negligible internal resistance is connected to two 10 Ω resistors in series.



What charge flows through each of the 10 Ω resistors in 1 minute?

A 0.30 C

B 0.60 C

C 3.0 C

D 18 C

Question 27 Medium [1 mark]

Two heating coils X and Y, of resistance R_X and R_Y respectively, deliver the same power when 12 V is applied across X and 6 V is applied across Y.

What is the ratio $\frac{R_X}{R_Y}$?

A $\frac{1}{4}$

B $\frac{1}{2}$

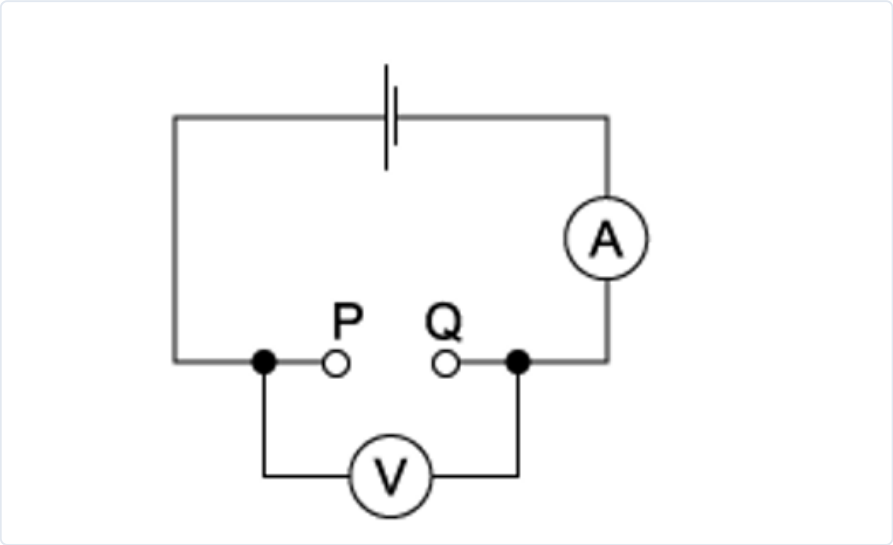
C 2

D 4

Question 28 Medium [1 mark]

A student found two unmarked resistors. To determine the resistance of the resistors, the circuit below was set up. The resistors were connected in turn between P and Q, noting the current readings.

The voltage readings were noted without the resistors and with each resistor in turn.



The results were entered into a spreadsheet, as shown.

1.5	1.3	28	46
1.5	1.4	14	100

The student forgot to enter the column headings.

Which order of the headings would be correct?

A	e.m.f / V	*V* / V	*R* / Ω	*I* / mA
B	*V* / V	e.m.f / V	*R* / Ω	*I* / mA
C	*V* / V	e.m.f / V	*I* / mA	*R* / Ω

D

e.m.f / V	*V* / V	*I* / mA	*R* / Ω

A

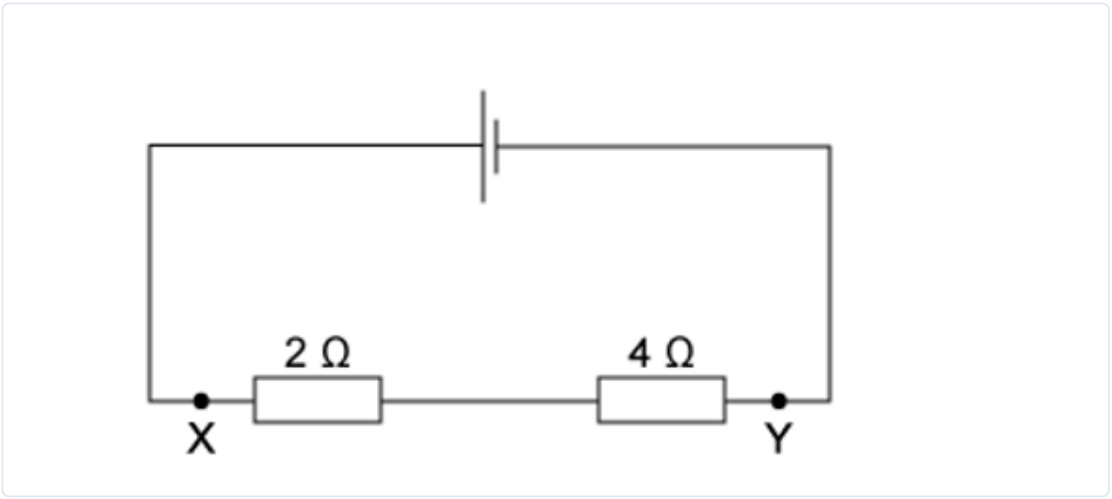
B

C

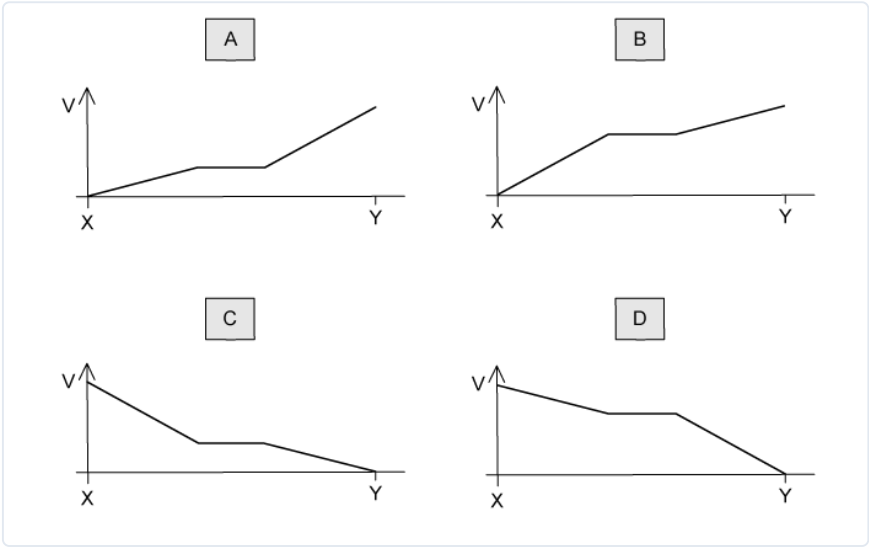
D

Question 29 Medium [1 mark]

A 2 Ω resistor and a 4 Ω resistor are connected to a cell.



Which graph shows how the potential V varies with distance between X and Y ?



A

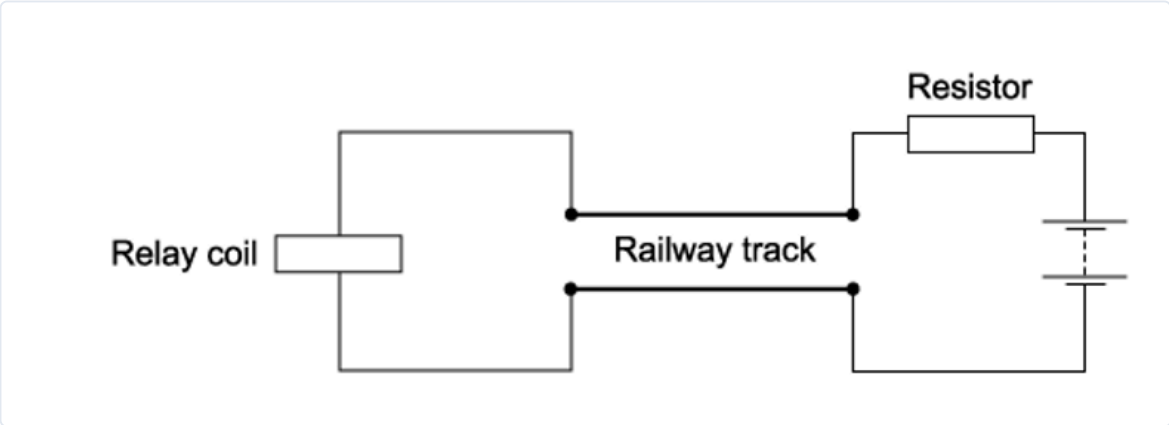
B

C

D

Question 30 Medium [1 mark]

The diagram shows a length of track from a model railway connected to a battery, a resistor and a relay coil.



With no train present, there is a current in the relay coil which operates a switch to turn on a light.

When a train occupies the section of track, most of the current flows through the wheels and axles of the train in preference to the relay coil. The switch in the relay turns off the light.

Why is a resistor placed between the battery and the track?

A to limit the heating of the wheels of the train

B to limit the energy lost in the relay coil when a train is present

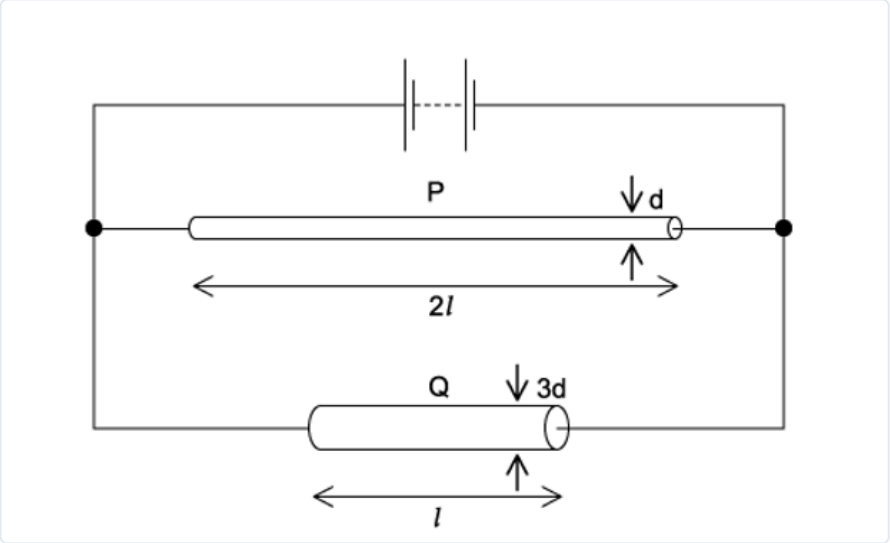
C to prevent a short circuit of the battery when a train is present

D to protect the relay when a train is present

Question 31 Medium [1 mark]

Two wires P and Q made of the same material are connected to the same electrical supply.

P has twice the length of Q and one-third of the diameter of Q, as shown in the diagram.



What is the ratio $\frac{current\ \ in\ \ P}{current\ \ in\ \ Q}$?

A $\frac{2}{3}$

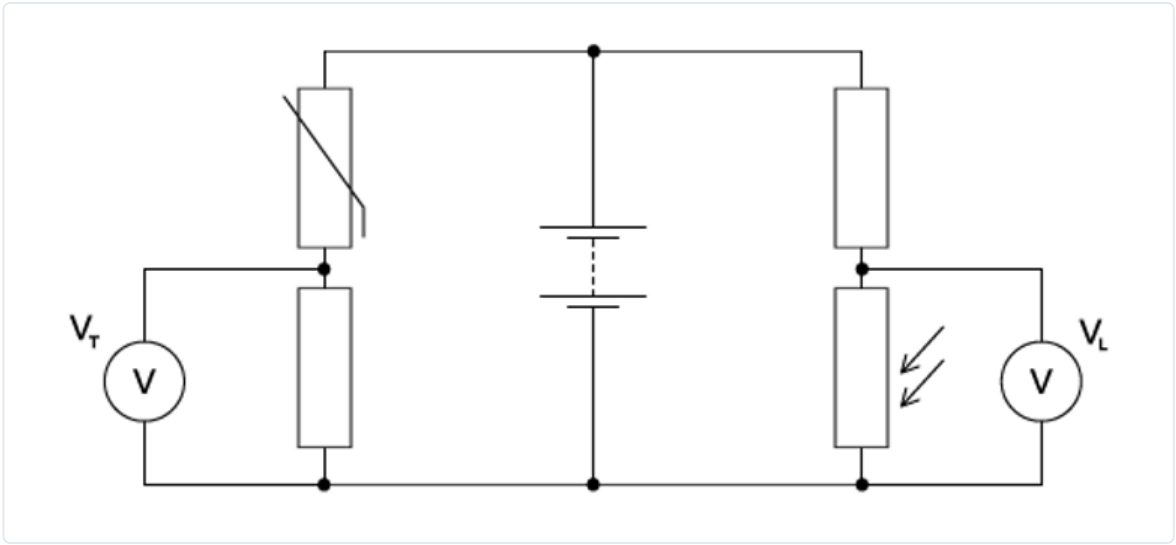
B $\frac{2}{9}$

C $\frac{1}{6}$

D $\frac{1}{18}$

Question 32 Medium [1 mark]

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



The readings V_T and V_L are both high.

What are the conditions of temperature and light level ?

	Temperature	Light level
A	low	low
B	low	high
C	high	low
D	high	high

A

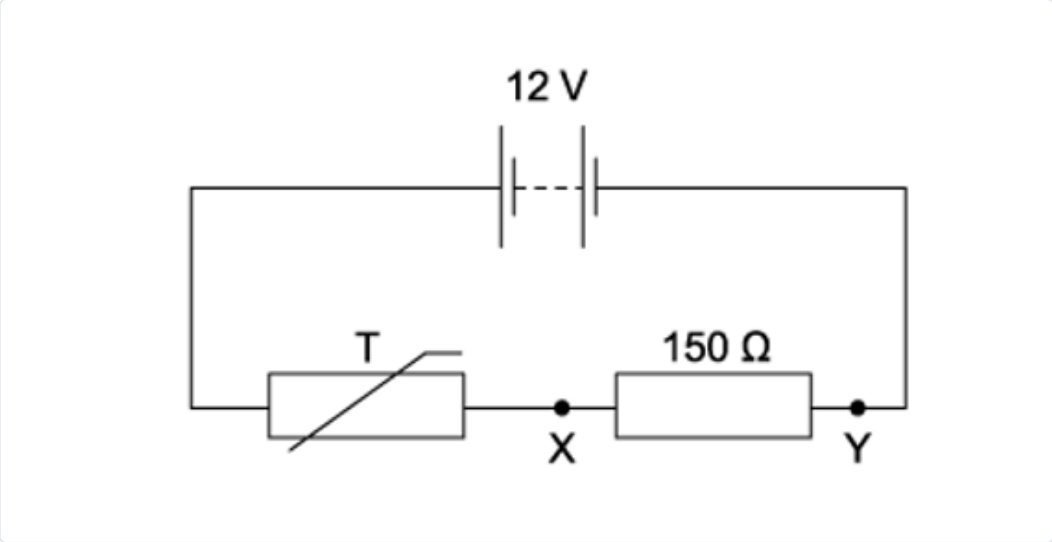
B

C

D

Question 33 Medium [1 mark]

In a fire alarm system, a thermistor T has a resistance of 2000 Ω at room temperature. Its resistance decreases as the temperature increases. The alarm is triggered when the potential difference between X and Y reaches 4.5 V.



What is the resistance of the thermistor when the alarm is triggered ?

A 90 Ω

B 150 Ω

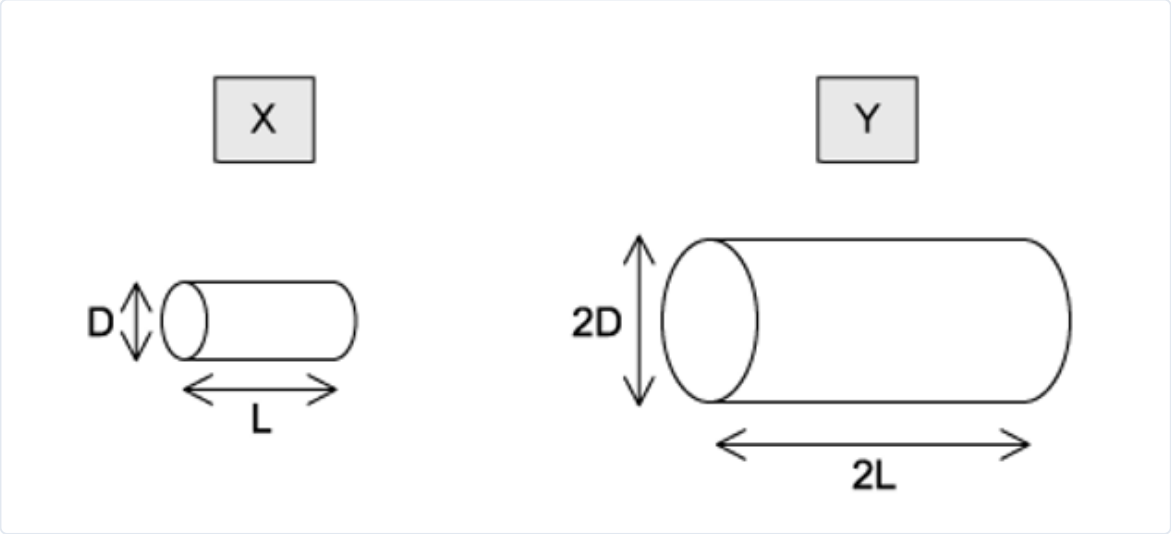
C 250 Ω

D 1300 Ω

Question 34 Medium [1 mark]

Two electrically-conducting cylinders X and Y are made from the same material.

Their dimensions are as shown.



The resistance between the ends of each cylinder is measured.

What is the ratio $\frac{resistance\ \ of\ \ X}{resistance\ \ of\ \ Y}$?

A $\frac{2}{1}$

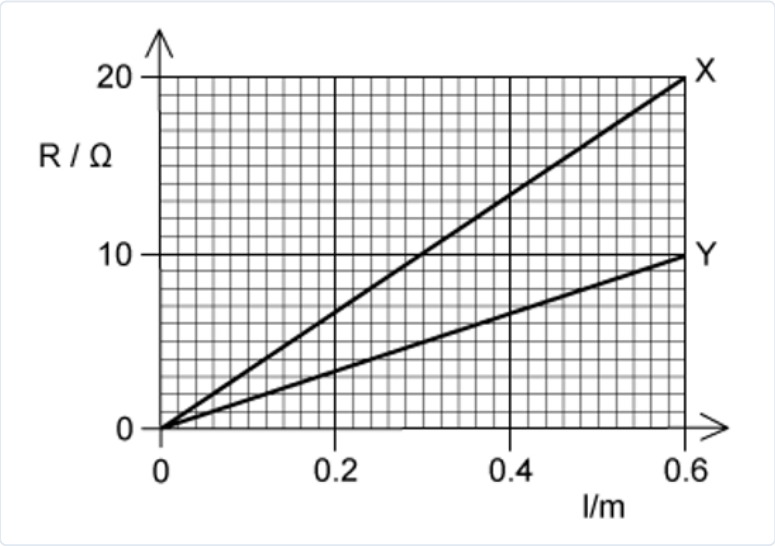
B $\frac{1}{1}$

C $\frac{1}{2}$

D $\frac{1}{4}$

Question 35 Medium [1 mark]

The graph shows the variation with length *l* of resistance *R* for two wires X and Y made from the same material.



What does the graph show?

A cross-sectional area of X = 2 × cross - sectional area of Y

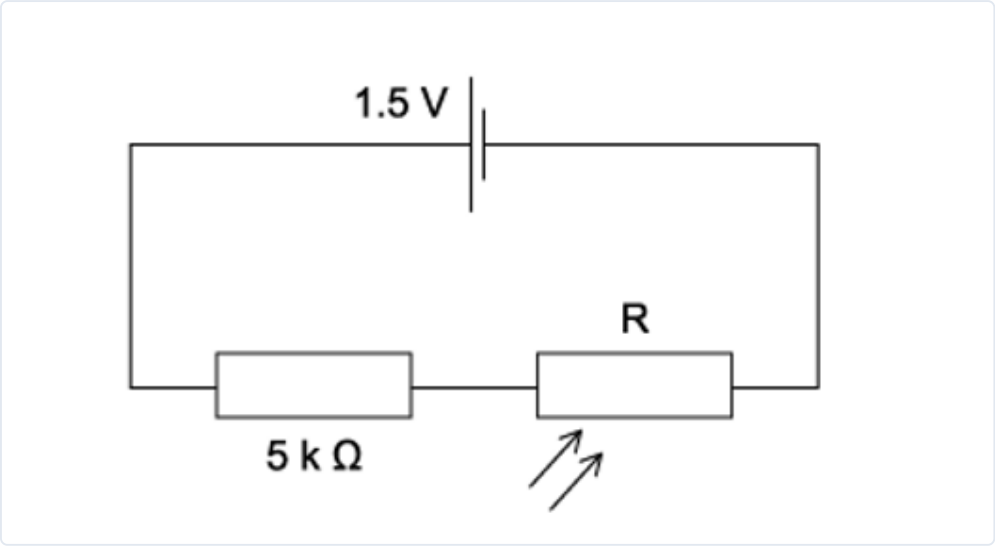
B resistivity of X = 2 × resistivity of Y

C when equal lengths of X and Y are connected in series to a battery, power in X = 2 × power in Y

D when equal lengths of X and Y are connected in parallel to a battery, current in X = 2 × current in Y

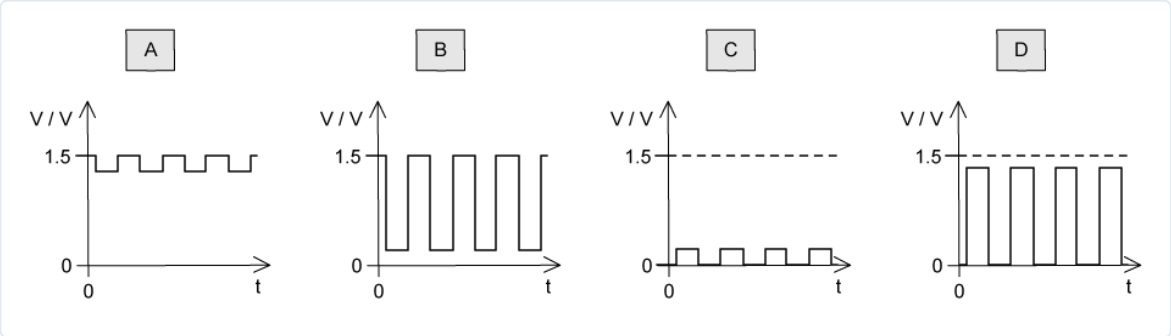
Question 36 Medium [1 mark]

A light-dependent resistor *R* has a resistance of about 1 MΩ in the dark and about 1 kΩ when illuminated. It is connected in series with a 5 kΩ resistor to a 1.5 V cell of negligible internal resistance.



The light-dependent resistor is illuminated (in an otherwise dark room) by a flashing light.

Which graph best shows the variation with time *t* of potential difference *V* across *R* ?



A

B

C

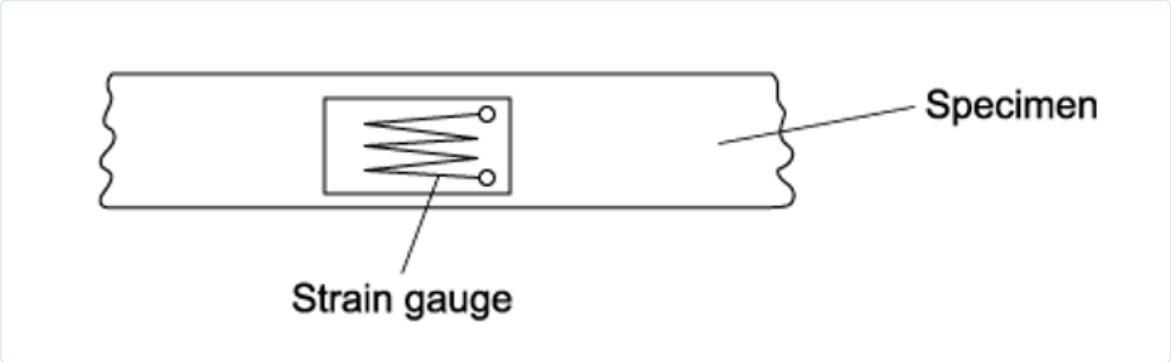
D

Question 37 Medium [1 mark]

Tensile strain may be measured by the change in electrical resistance of a strain gauge.

A strain gauge consists of folded fine metal wire mounted on a flexible insulating backing sheet.

The strain gauge is firmly attached to the specimen, so that the strain in the metal wire is always identical to that in the specimen.



When the strain in the specimen is increased, what happens to the resistance of the wire?

A it decreases, because the length decreases and the cross-sectional area increases

B it decreases, because the length increases and the cross-sectional area decreases

C it increases, because the length decreases and the cross-sectional area increases

D it increases, because the length increases and the cross-sectional area decreases

Question 38 Medium [1 mark]

The resistance of a thermistor depends on its temperature, and the resistance of a light-dependent resistor (LDR) depends on the illumination.

Under which conditions will the resistance of both a thermistor and an LDR be highest ?

	Thermistor	LDR
A	highest temperature	highest illumination
B	highest temperature	lowest illumination
C	lowest temperature	highest illumination
D	lowest temperature	lowest illumination

A

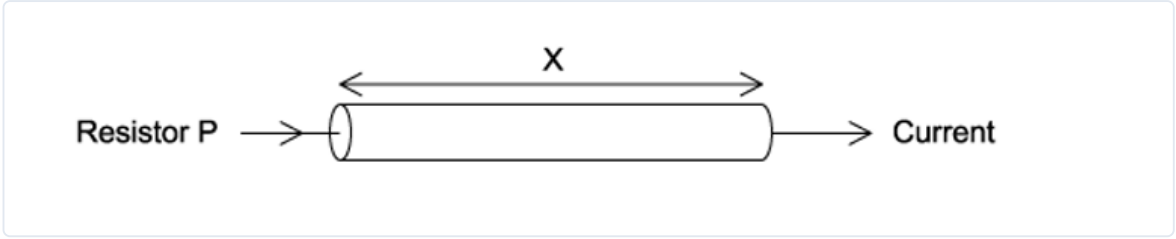
B

C

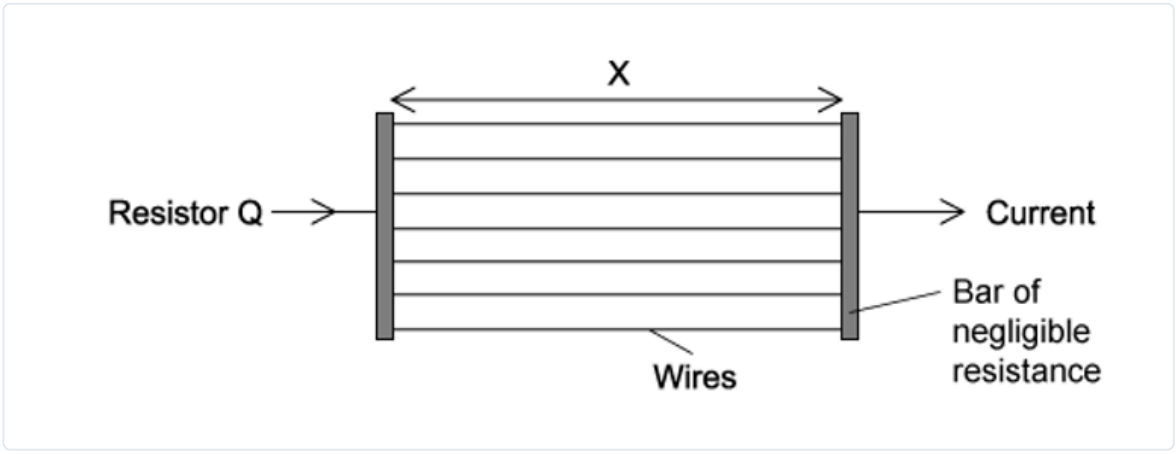
D

Question 39 Medium [1 mark]

A researcher has two pieces of copper of the same volume. All of the first piece is made into a cylindrical resistor P of length *x*.



All of the second piece is made into uniform wires each of the same length *x* which he connects between two bars of negligible resistance to form a resistor Q.



How do the electrical resistances of P and Q compare ?

A P has a larger resistance than Q

B Q has a larger resistance than P

C P and Q have equal resistance

D Q may have a larger or smaller resistance than P, depending on the number of wires made

Question 40 Medium [1 mark]

Two wires have the same length and the same resistance. Wire X is made of a metal of resistivity 1.7×10^{-8} Ωm, and wire Y is made of a metal of resistivity 5.6×10^{-8} Ωm.

The diameter of wire X is 0.315 mm.

What is the diameter of wire Y?

A 0.17 mm

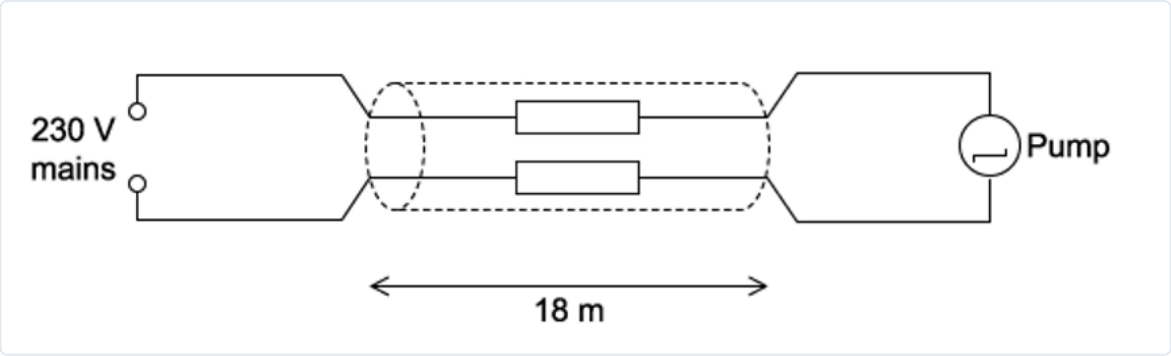
B 0.33 mm

C 0.57 mm

D 1.0 mm

Question 41 Hard [1 mark]

The diagram shows an electric pump for a garden fountain connected by an 18 m cable to a 230 V mains electrical supply.



The performance of the pump is acceptable if the potential difference (p.d.) across it is at least 218 V. The current through it is then 0.83 A.

What is the maximum resistance per meter of each of the two wires in the cable if the pump is to perform acceptably?

A $0.40\ \Omega\ m^{-1}$

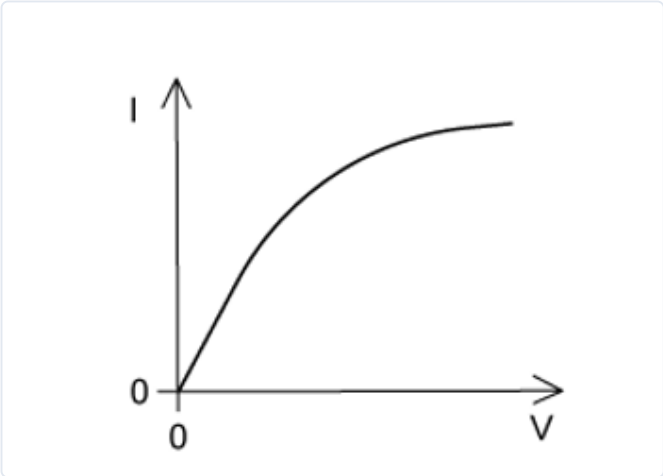
B $0.80\ \Omega\ m^{-1}$

C $1.3\ \Omega\ m^{-1}$

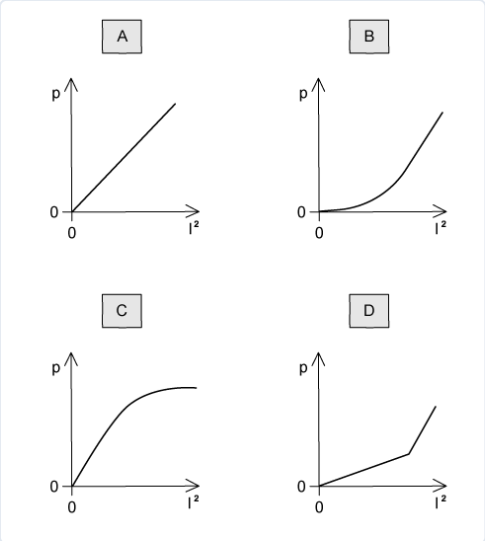
D $1.4\ \Omega\ m^{-1}$

Question 42 Hard [1 mark]

The graph below shows the variation of the current *I* through a lamp with potential difference *V* across it.



Which of the following graphs best represents the variation of power *P* dissipated in the same lamp with I^2?



A

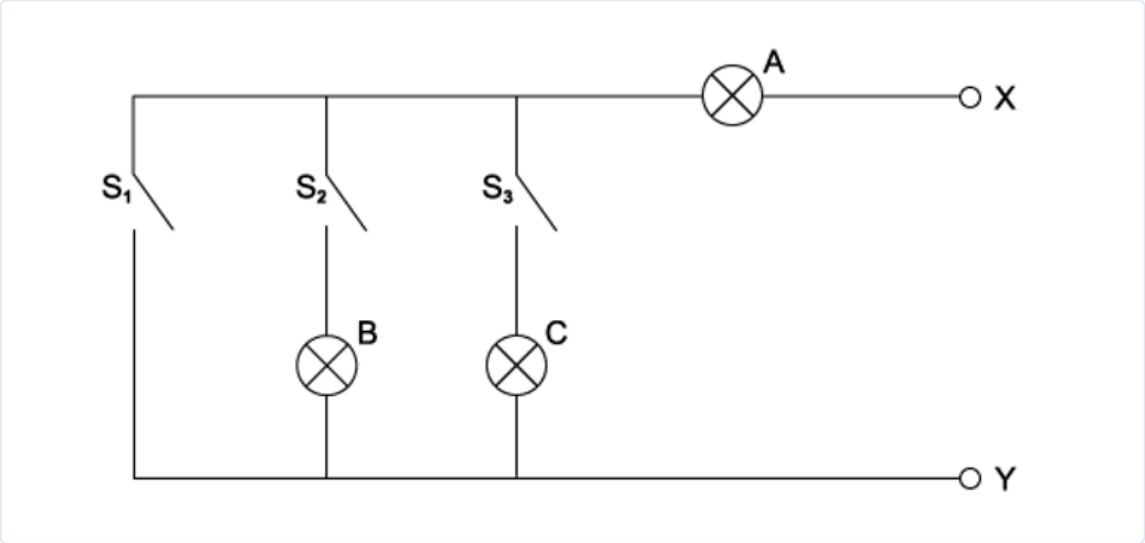
B

C

D

Question 43 Hard [1 mark]

Circuit contains three identical lamps **A**, **B** and **C**. Circuit also contains three switches, S_1, S_2 and S_3, as shown.



One of the lamps is faulty. In order to detect the fault, an ohm-meter is connected between terminals X and Y.

When measuring resistance, the ohm-meter causes negligible current in the circuit. Readings of the ohm-meter for different switch positions are shown in the table.

Switch			Meter reading / Ω
S_1	S_2	S_3	
open	open	open	∞
closed	open	open	15
open	closed	open	30
open	closed	closed	15

Which lamps are faulty?

A A & B

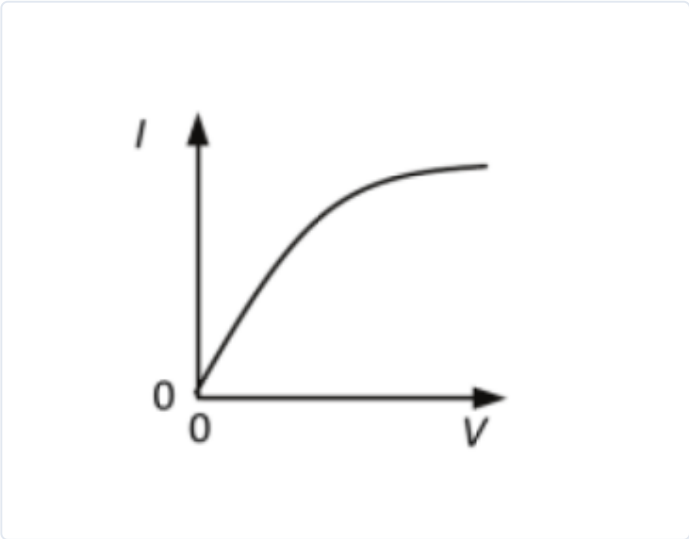
B B only

C B & C

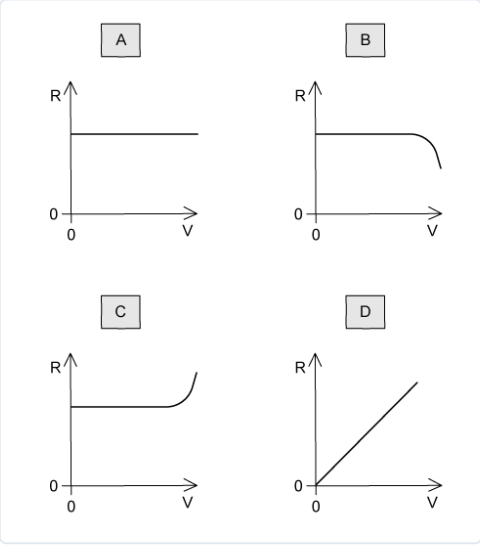
D C only

Question 44 Hard [1 mark]

The current flowing through a component varies with the potential difference *V* across it as shown.



Which graph best represents how the resistance *R* varies with *V* ?



A

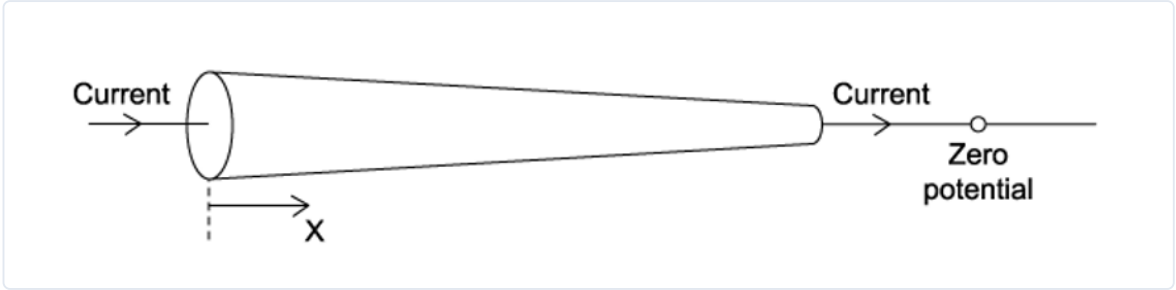
B

C

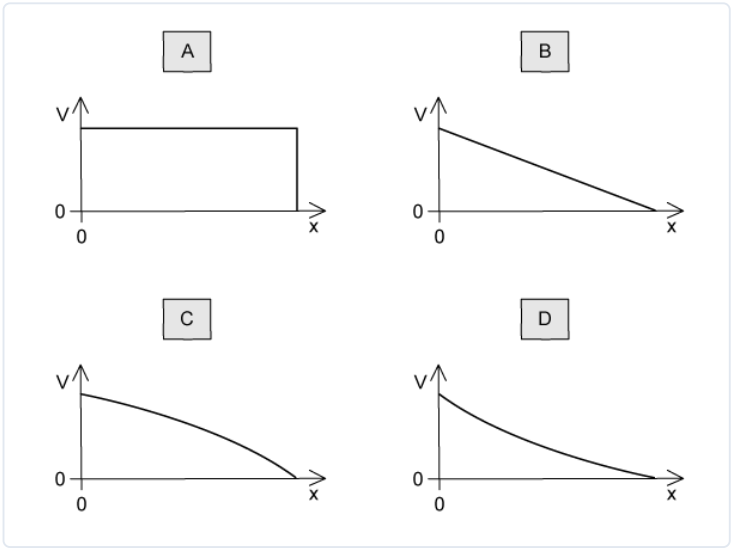
D

Question 45 Hard [1 mark]

The circular cross-sectional area of a uniform metal wire varies along its length. There is a current in the wire. The narrow end of the wire is at a reference potential of zero.



Which graph best represents the variation with distance *x* along the wire of the potential difference *V* relative to the reference zero?



A

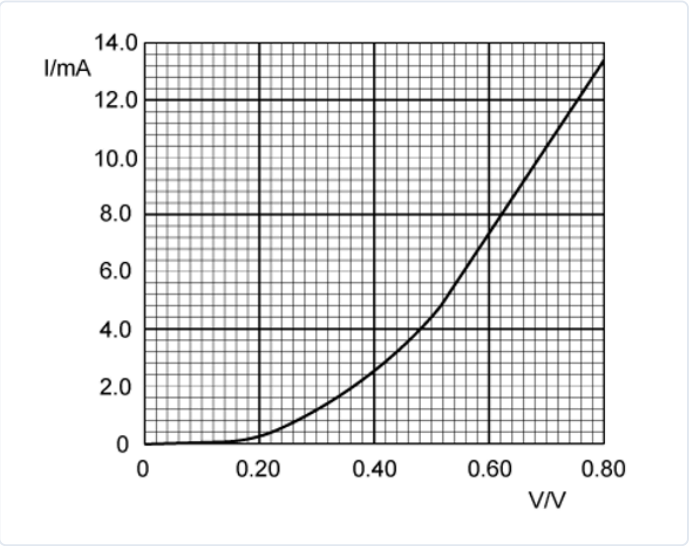
B

C

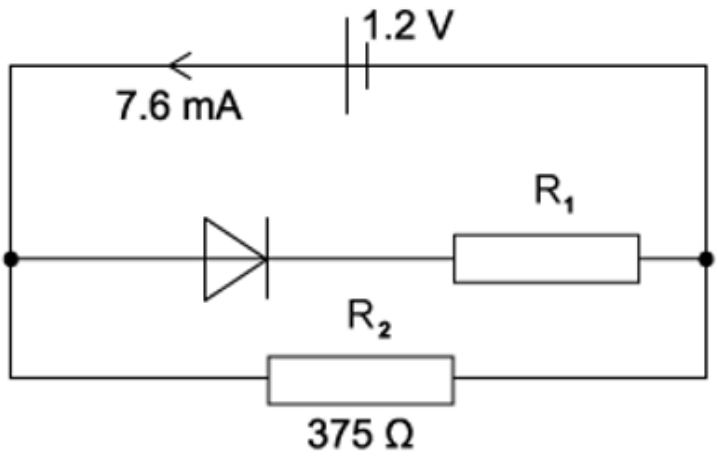
D

Question 46 Hard [1 mark]

The graph shows the *I* -*V* characteristics of a semiconductor diode



A cell of e. m .f. 1.2 V and negligible internal resistance is connected in series to a semiconductor diode and a resistor R_1, as shown below



A resistor R_2 of resistance 375 Ω is connected across the cell. The diode has the characteristic shown in the graph. The current supplied by the cell is 7.6 mA.

What is the resistance of R_1?

A 82 Ω

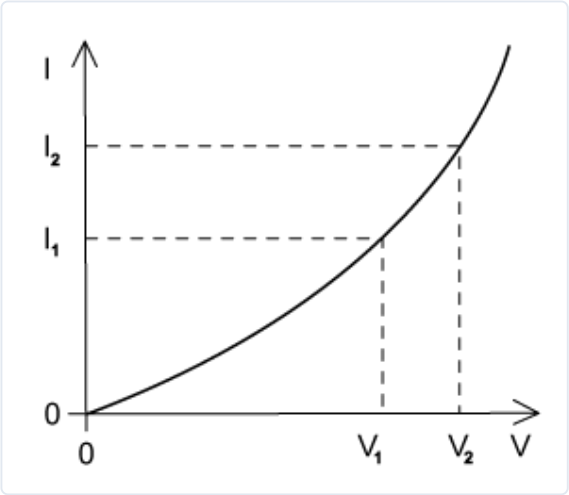
B 114 Ω

C 160 Ω

D 260 Ω

Question 47 Hard [1 mark]

The graph shows how the current *I* varies with potential difference *V* for an electrical component.



Which row of the table gives the resistance of the component at V_2 and describes how the resistance changes from V_1 to V_2?

	Resistance at V_2	Change in resistance from V_1 to V_2
A	$\frac{V_2 - V_1}{I_2 - I_1}$	increases
B	$\frac{V_2 - V_1}{I_2 - I_1}$	decreases
C	$\frac{V_2}{I_2}$	increases
D	$\frac{V_2}{I_2}$	decreases

A

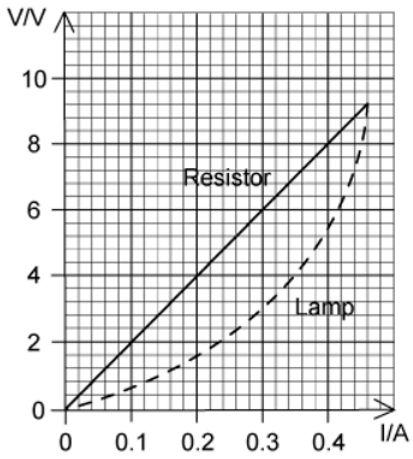
B

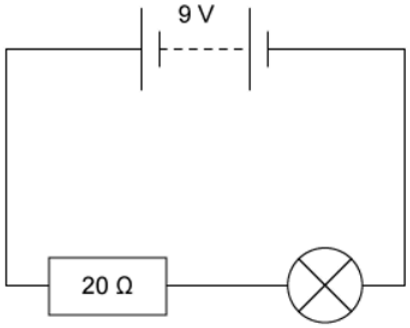
C

D

Question 48 Hard [1 mark]

The graph shows the relationship between potential difference *V* and current *I* for a fixed 20 Ω resistor and a filament lamp.





The resistor and lamp are placed in series with a 9 V battery of negligible internal resistance. What is the current in the circuit ?

A 0.1 A

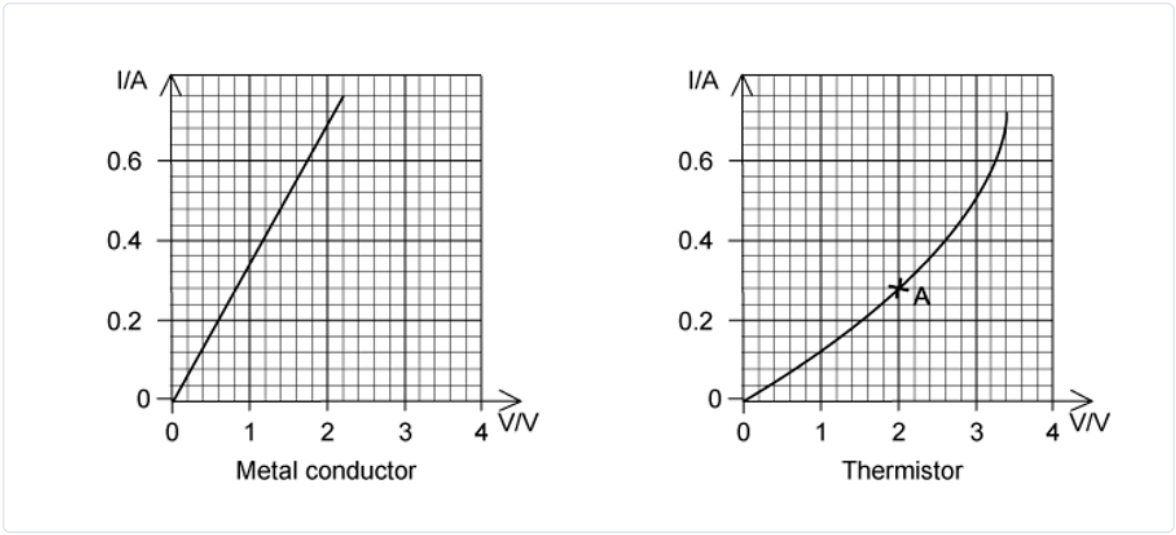
B 0.2 A

C 0.3 A

D 0.4 A

Question 49 Hard [1 mark]

The graphs show the *I* -*V* characteristics for a metal conductor and a thermistor



What is the ratio $\frac{resistance\ \ of\ \ thermistor\ \ at\ \ point\ \ A}{resistance\ \ of\ \ metal\ \ conductor}$?

A $\frac{2}{5}$

B $\frac{1}{2}$

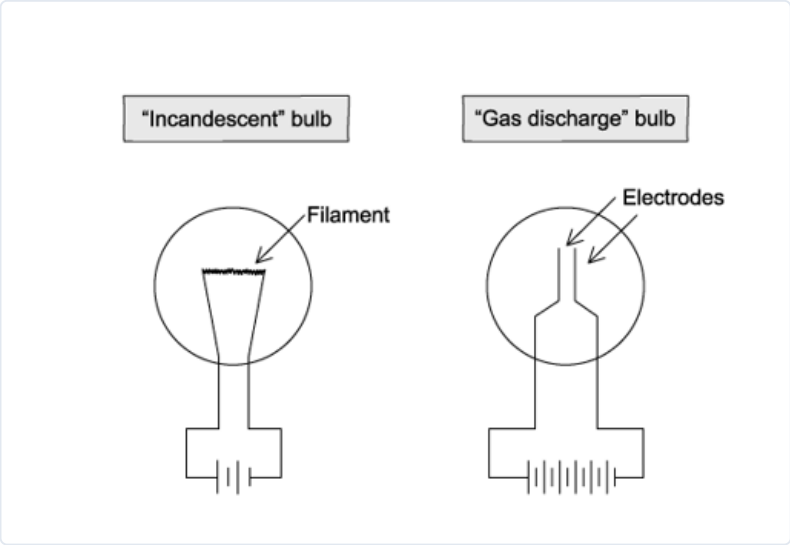
C $\frac{2}{1}$

D $\frac{5}{2}$

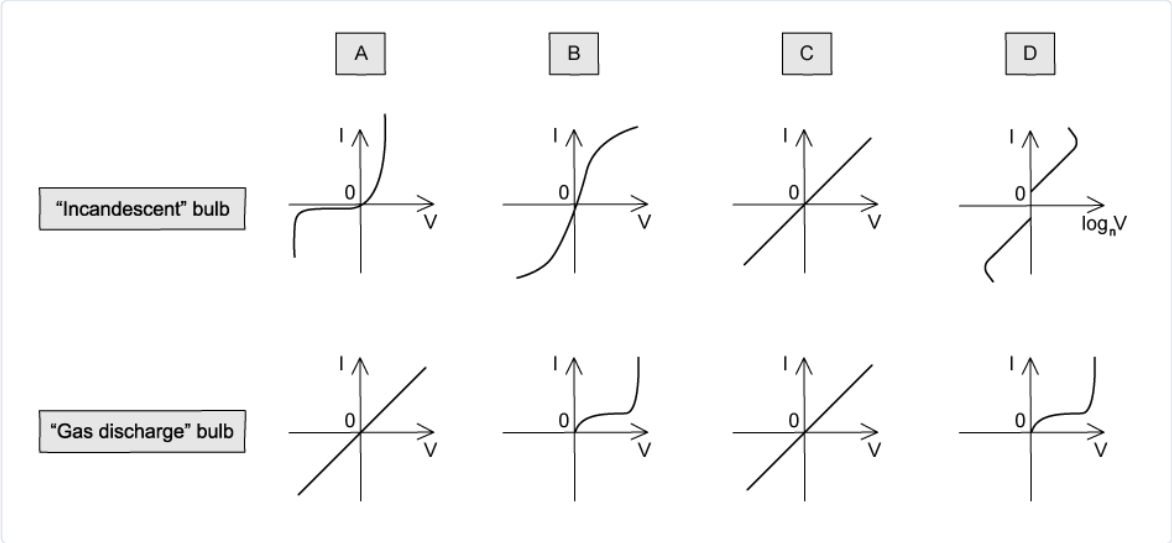
Question 50 Hard [1 mark]

One style of light bulb, very different from the "incandescent" design which works on the principle of a super-heated wire filament emitting light, is called a gas discharge tube.

In this design of light bulb, light is produced by the direct excitation of gas molecules as electric current passes between two electrodes.



What will the *I* - *V* graphs of these bulbs look like?



A

B

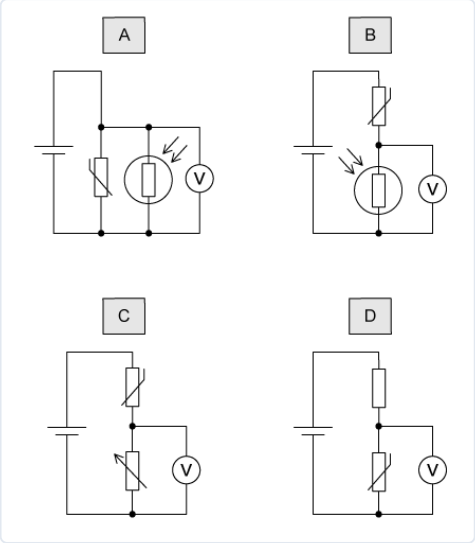
C

D

Question 51 Hard [1 mark]

A thermistor and another component are connected to a constant voltage supply. A voltmeter is connected across one of the components. The temperature of the thermistor is then reduced, but no other changes are made.

In which circuit will the voltmeter reading increase ?



A

B

C

D

Question 52 Hard [1 mark]

A copper wire is to be replaced by an aluminum alloy wire of the same length and resistance.

Copper has half the resistivity of the alloy.

What is the ratio $\frac{diameter\ \ of\ \ alloy\ \ wire}{diameter\ \ of\ \ copper\ \ wire}$?

A $\sqrt{2}$

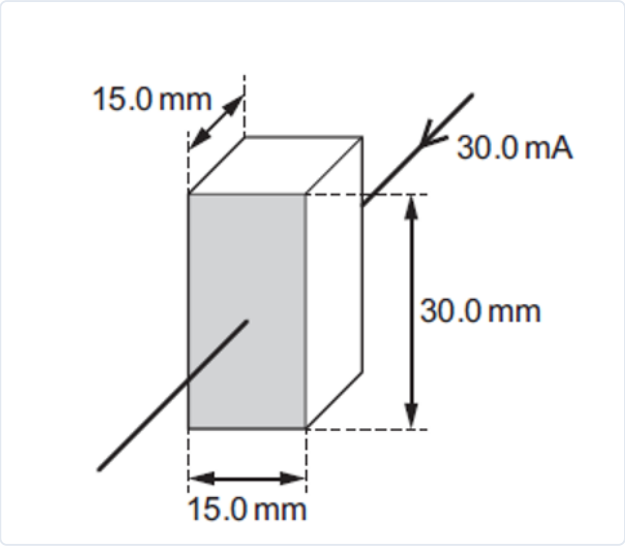
B 2

C $\sqrt[3]{2}$

D 4

Question 53 Hard [1 mark]

The current in a block of semiconductor is 30.0 mA when there is a potential difference (p.d.) of 10.0 V across it. The dimensions of the block and the direction of the current in it are as shown.



The electrical meters used are accurate to ± 0.1 mA and ± 0.1 V. The dimensions of the block are accurate to ± 0.2 mm.

What is the resistivity of the semiconductor?

A 10.0 ± 0.2 Ω m

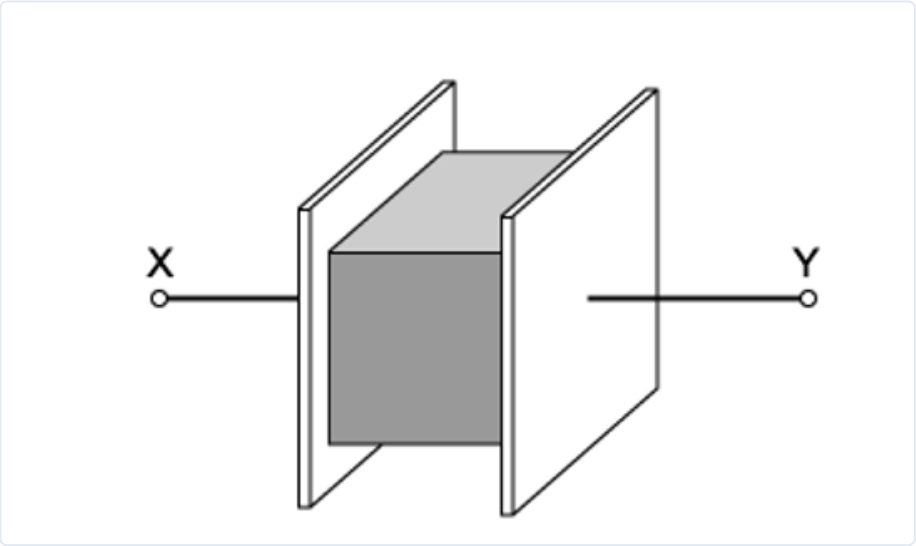
B 10.0 ± 0.3 Ω m

C 10.0 ± 0.5 Ω m

D 10.0 ± 0.8 Ω m

Question 54 Hard [1 mark]

The resistance of a metal cube is measured by placing it between two parallel plates, as shown.



The cube has volume V and is made of a material with resistivity ρ. The connections to the cube have negligible resistance.

Which expression gives the electrical resistance of the metal cube between X and Y?

A $\rho V^{\frac{1}{3}}$

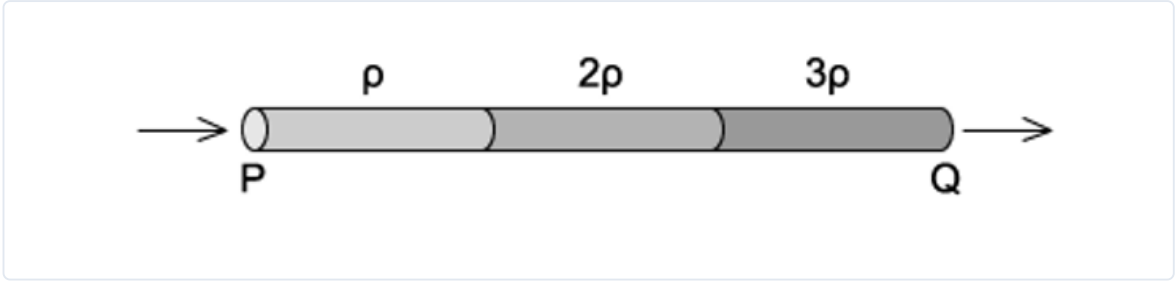
B $\rho V^{\frac{2}{3}}$

C $\frac{\rho}{V^{\frac{1}{3}}}$

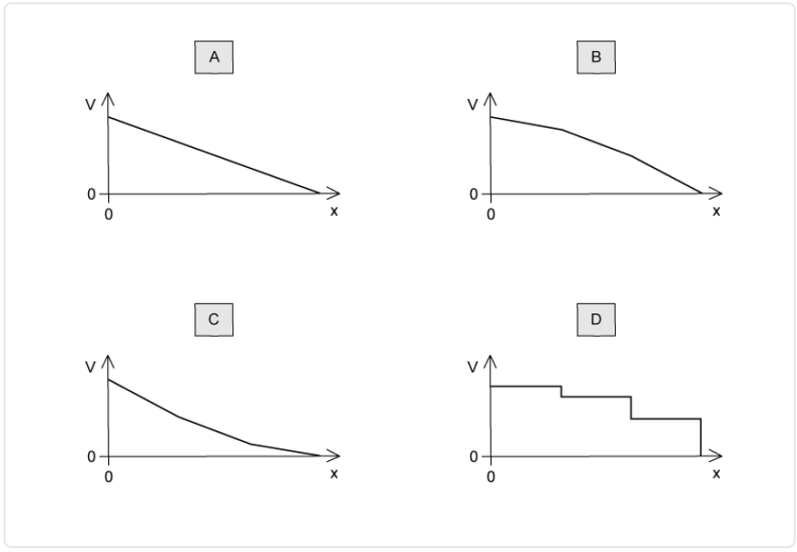
D $\frac{\rho}{V^{\frac{2}{3}}}$

Question 55 Hard [1 mark]

A wire PQ is made of three different materials, with resistivities ρ, 2ρ and 3ρ. There is a current I in this composite wire, as shown.



Which graph best shows how the potential V along the wire varies with distance x from P ?



A

B

C

D

Question 56 Hard [1 mark]

A strain gauge consists of a length of wire with a uniform cross-sectional area. Its resistance is 2.50 kΩ. It is attached to a large pipe.

When the pipe expands, the strain gauge changes in dimensions, such that the length increases by 4% and the diameter reduced by 1%.

What is the new resistance of the strain gauge?

A 2.60 kΩ

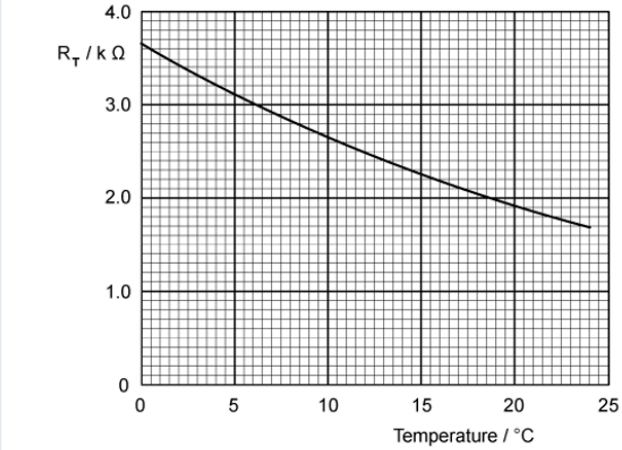
B 2.65 kΩ

C 2.80 kΩ

D 3.00 kΩ

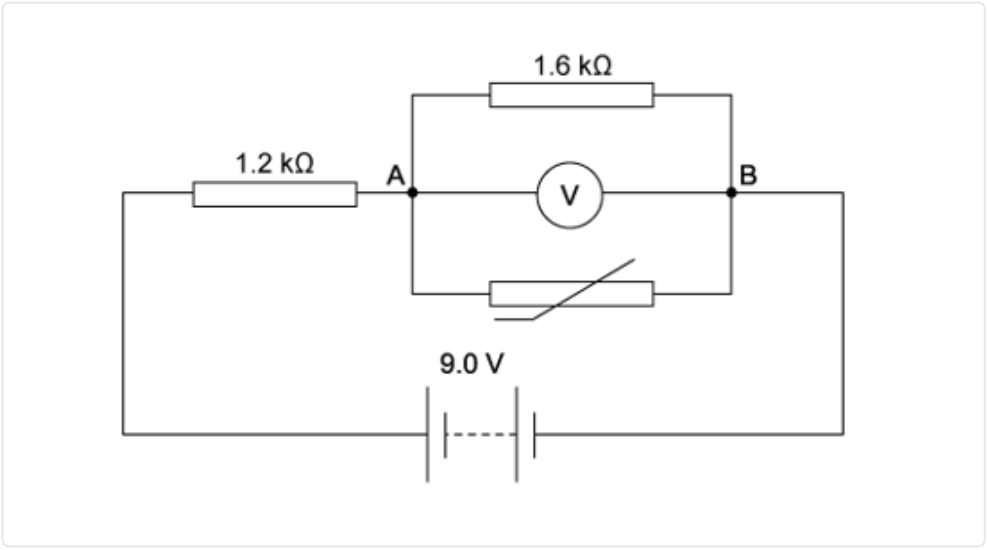
Question 57 Hard [1 mark]

The variation with temperature of the resistance R_T of a thermistor is shown on the graph.



The thermistor is connected into the circuit shown below.

The battery has an e. m .f. of 9.0 V and negligible internal resistance. The voltmeter has infinite resistance.



What is the temperature of the thermistor when the reading on the voltmeter is 4.0 V?

A 2.5 °C

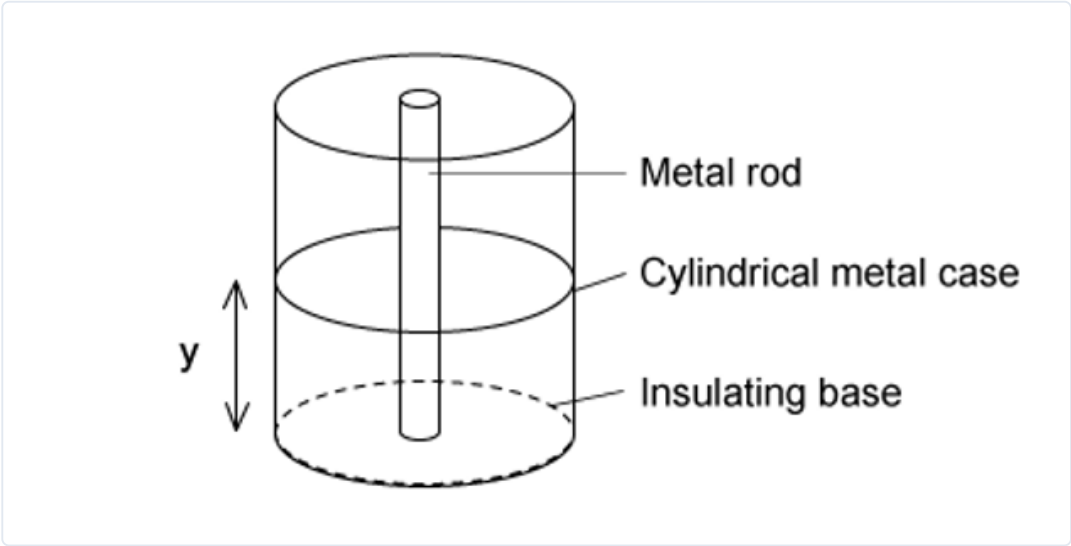
B 13.0 °C

C 16.0 °C

D 24.0 °C

Question 58 Hard [1 mark]

A conducting liquid fills a cylinder metal case to a depth *y* as shown in the diagram.



The resistance between the case and the metal rod is

A proportional to *y*

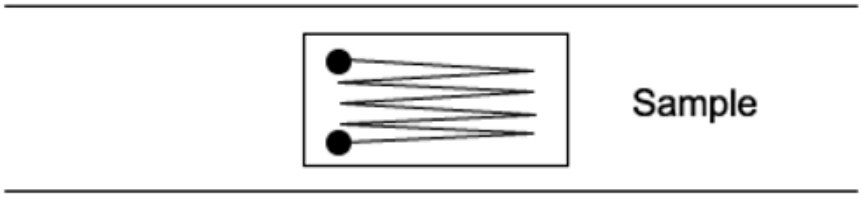
B proportional to y^2

C inversely proportional to *y*

D inversely proportional to y^2

Question 59 Hard [1 mark]

The following strain gauge consists of a fine metal wire, so that an applied tension on the strain gauge results in changes in the length and cross-sectional area of the wire.



When the length of the wire is stretched by 5%, what is the percentage change in the resistance of the wire? Assume that the volume of a wire remains constant when it is stretched.

A + 5%

B – 5%

C + 10%

D – 10%

Question 60 Hard [1 mark]

Aluminum and copper rods are designed to have the same length and the same resistance.

The resistivity of copper is half that of aluminum, but its density is three times that of aluminum.

Determine the ratio of the mass of the aluminum rod to the mass of the copper rod.

A $\frac{1}{6}$

B $\frac{2}{3}$

C $\frac{3}{2}$

D $\frac{6}{1}$